COMSATS University Islamabad
Department of Computer Science

Course Information	
Course Code: **CSC354**	Course Title: **Machine Learning**
Credit Hours: **3(3,0)**	Lecture Hours/Week: **3**
Lab Hours/Week: **0**	Pre-Requisites: **None**

Text and Reference Books

Textbooks:

1. Introduction to Machine Learning, Ethem Alpaydin, MIT Press, 2010.
2. Machine Learning, Tom, M., McGraw Hill, 1997.

Reference Books:

1. Hands on Machine Learning with Scikit-Learn and TensorFlow, Aurelien Geron, O`Reilly Media, 2017.
2. Deep Learning with PyTorch – Essential Excerpts, Eli Stevens, Luca Antiga, Thomas Viehmann, Manning Publications, 2009.
3. Pattern Recognition and Machine Learning, Bishop, C., Springer-Verlag, 2007.

- Zero tolerance for any misconduct and misbehavior
- Zero tolerance for cheating, if student found with any cheating material in quiz/exam will be marked zero
- Copied assignment will be marked zero
- Cross talk with class-fellows will result in strict disciplinary action
- Be in class on time, students who arrive after 5 min will be marked absent
- Submit assignments on portal on time, late assignment wouldn't be considered, and anything send via (WhatsApp etc) wouldn't be considered
- Emails or any message should be sent within working hours rest would not be entertained send email on **(zeenatzulfiqar@cuiatd.edu.pk)**

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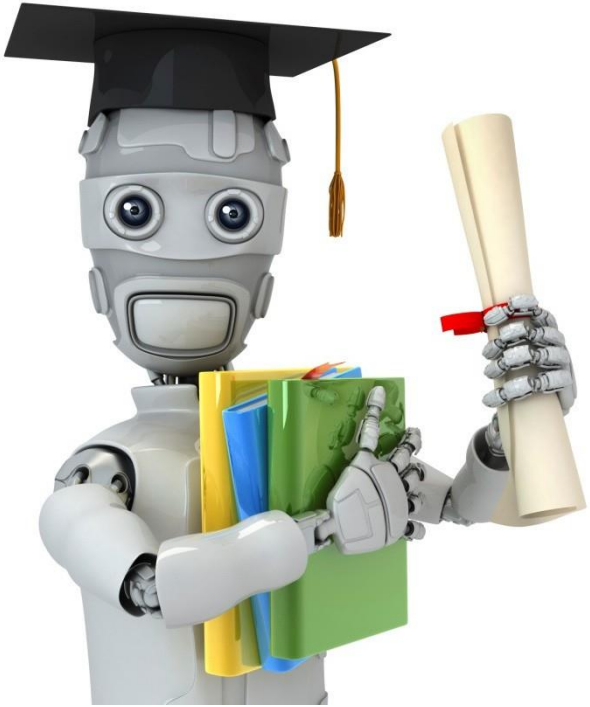


Machine Learning

Introduction

Decision Tree

Decision Tree

1. Supervised Learning Algorithm

•Used for both **classification and regression** problems.

2. Tree Structure

Consists of:

- **Root Node** (starting point)
- **Internal Nodes** (decision points)
- **Leaf Nodes** (final output)

3. Recursive Splitting

Data is divided step-by-step into smaller subsets based on conditions.

Decision Tree

- **Easy to Understand**

Works like human decision-making **(if-else rules).**

- **Handles Both Data Types**

Can work with **numerical and categorical data**.

- **No Need for Data Scaling**

Does not require normalization or standardization.

- **Overfitting Problem**

Can memorize training data if the tree becomes too deep.

- **Non-Parametric Model**
- Does not assume any fixed form of data distribution.
- **Fast Prediction**
- Once trained, it makes predictions quickly.
- **Sensitive to Data Changes**
- Small changes in data can create a completely different tree.

Decision Tree Terminologies

Root Node: Root node is from where the decision tree starts. It represents the entire dataset, which further gets divided into **two or more homogeneous sets.**

- **Leaf Node:** Leaf nodes are the final output node, and the tree cannot be segregated further after getting a leaf node.
- **Splitting:** Splitting is the process of dividing the decision node/root node into sub-nodes according to the given conditions.
- **Branch/Sub Tree:** A tree formed by splitting the tree.
- **Pruning:** Pruning is the process of removing the unwanted branches from the tree.
- **Parent/Child node:** The root node of the tree is called the parent node, and other nodes are called the child nodes

Decision Tree Types

Univariate Trees
- Classification Trees
- Regression Trees

Univariate Decision Trees

- It can be used for both **classification and regression problems**.
- The structure is **simple and easy to interpret** (if–else rules).
- It is **computationally efficient** and faster to train.
- Works well with both **numerical and categorical data**.
- Cannot capture **interaction between multiple features** at the same time.
- May lead to **lower accuracy** for complex datasets.
- Prone to **overfitting** if the tree becomes too deep.
- **Pruning techniques** are used to control overfitting.

Univariate Decision Trees

- Each decision node applies a **single condition** (e.g., Age < 30).
- It divides the dataset into **smaller and more homogeneous subsets**.
- A **Univariate Decision Tree** uses **only one feature (variable)** at each split.
- The model selects the best feature based on criteria like:

  - **Information Gain**
  - **Gini Index**
  - **Variance Reduction**

Classification Tree

- A **Classification Tree** is a type of **decision tree** used when the **target variable is categorical** (e.g., **Yes/No**, **Pass/Fail**).
- It classifies data into **different classes or categories** based on **input features.**
- The **tree is built** using a **top-down recursive splitting approach**.
- At each node, the best feature is selected to **separate the classes effectively**.
- **Common splitting criteria include:**
- **Gini Index**
- **Entropy (Information Gain)**
- The **goal is** to create **pure nodes** (nodes containing data **from only one class**).
- The **final output** is shown at **leaf nodes**, which **represent class labels**.
- It works like a set of **if–else decision rules**.

Classification Tree

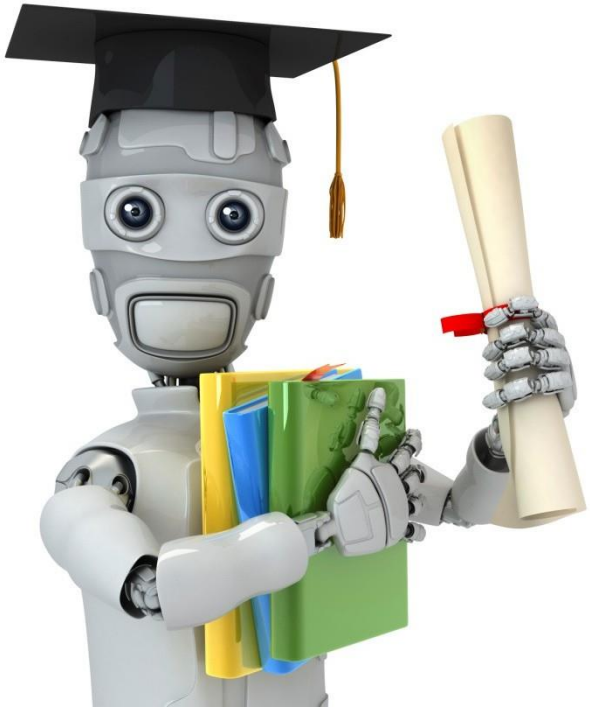
- It can handle both **categorical and numerical input features**.
- It is **easy to understand and interpret**, even for **non-technical users.**
- It does **not require data normalization or scaling**.
- It may suffer from **overfitting** if the tree becomes too complex.
- **Pruning** is used to reduce overfitting and improve generalization.
- It is sensitive to **small changes in data**, which may produce a different tree.

Regression Tree

- It predicts **numerical values** instead of **class labels**.
- A **Regression Tree** is a type of decision tree used when the **target variable is continuous (numeric)**.
- The dataset is split into smaller subsets based on **feature conditions** (e.g., X < 5).
- It uses a **top-down recursive splitting approach**.
- The **goal is to minimize** prediction error in each split.
- **Common splitting criteria include:**
- **Mean Squared Error (MSE)**
- **Variance Reduction**
- Each leaf node contains the **average (mean) value** of the target variable in that subset.
- It can handle both **numerical and categorical input features**.

Regression Tree

- It captures **nonlinear relationships** between variables.
- It is **easy to interpret** using if–else rules.
- No need for **data normalization or scaling**.
- **Pruning techniques** are used to reduce overfitting.
- Small changes in data can result in a **different tree structure**.
- It may suffer from **overfitting** if the tree becomes too deep.



ID3 (Iterative Dichotomiser 3) is a **decision tree learning algorithm** used in machine learning for **classification problems**.

How ID3 works

1. Start with the **full dataset.**
2. Calculate **entropy (uncertainty)** of the dataset.
3. For each attribute, calculate **information gain**.
4. Select the attribute with **maximum information gain**.
5. Split the dataset based on that attribute.
6. Repeat the process for each branch.
7. Stop when:
    1. All data in a node belongs to one class, OR
    2. No attributes are left.

Instance	a_1	a_2	Target Class
1	T	T	+
2	T	T	+
3	T	F	−
4	F	F	+
5	F	T	−
6	F	T	−
7	F	F	−
8	T	F	+
9	F	T	−

Classification Tree Problem

- How to find the Entropy
- Information Gain
- Gini Index
- Splitting Attributes

How to find the Entropy

Step 1: Count Classes

- Positive (+) = 4
- Negative (−) = 5
- Total = 9

Step 3: Entropy of Full Dataset

$$Entropy(S) = -p_+ \log_2(p_+) - p_- \log_2(p_-)$$

Probabilities:

- $p_+ = 4/9$
- $p_- = 5/9$

Calculation:

$$Entropy(S) = -\frac{4}{9}\log_2\left(\frac{4}{9}\right) - \frac{5}{9}\log_2\left(\frac{5}{9}\right)$$

Entropy(S) ≈ **0.991 bits**

Instance	a_1	a_2	Target Class
1	T	T	+
2	T	T	+
3	T	F	−
4	F	F	+
5	F	T	−
6	F	T	−
7	F	F	−
8	T	F	+
9	F	T	−

GINI Index of Full Dataset

Formula:

$$Gini = 1 - (p_+^2 + p_-^2)$$

Step 1: Substitute

$$Gini = 1 - \left(\left(\tfrac{4}{9}\right)^2 + \left(\tfrac{5}{9}\right)^2\right)$$

Gini ≈ 0.494

Information Gain for a1

We divide the dataset into two groups:

Given Split on a1

We divide the dataset into two groups:

Group 1: a1 = T (True)

Instances: 1, 2, 3, 8

Class distribution:

•Positive (+) = 3

•Negative (−) = 1

So this subset is:

$$(3^{+}, 1^{-})$$

Instance	a_1	a_2	Target Class
1	T	T	+
2	T	T	+
3	T	F	−
4	F	F	+
5	F	T	−
6	F	T	−
7	F	F	−
8	T	F	+
9	F	T	−

Information Gain

Group 2: a1 = F (False)
Instances: 4, 5, 6, 7, 9
Class distribution:
•Positive (+) = 1
•Negative (−) = 4
So this subset is:

$$(1^{+}, 4^{-})$$

Instance	a_1	a_2	Target Class
1	T	T	+
2	T	T	+
3	T	F	−
4	F	F	+
5	F	T	−
6	F	T	−
7	F	F	−
8	T	F	+
9	F	T	−

Entropy for a1

For a1 = T:

$$Entropy(T) = -\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right)$$

→ ≈ 0.811

For a1 = F:

$$Entropy(F) = -\frac{1}{5}\log_2\left(\frac{1}{5}\right) - \frac{4}{5}\log_2\left(\frac{4}{5}\right)$$

→ ≈ 0.722

Weighted Entropy

$$Entropy_{split} = \frac{|S_1|}{|S|} \times Entropy(S_1) + \frac{|S_2|}{|S|} \times Entropy(S_2)$$

Substitute values:

$$= \frac{4}{9} \times 0.811 + \frac{5}{9} \times 0.722$$

Entropy(a1) ≈ 0.761

Information Gain (a1)

G(a1)=Entropy(S)−Entropy(a1)

$$IG(a1) = 0.991 - 0.761$$

IG(a1) ≈ 0.230

where:

•**Entropy(S)** = impurity of full dataset (before split)

•**Entropy(a1)** = weighted entropy after splitting on a1

Split on Attribute a2

a2 = T
Instances: 1, 2, 5, 6, 9
Class distribution:
•**Positive** (+) = 2
•**Negative** (−) = 3
So: $(2^{+}, 3^{-})$
Total = 5 instances
a2 = F
Instances: 3, 4, 7, 8
Class distribution:
•Positive (+) = 2
•Negative (−) = 2
So: $(2^{+}, 2^{-})$
Total = 4 instances

Instance	a_1	a_2	Target Class
1	T	T	+
2	T	T	+
3	T	F	−
4	F	F	+
5	F	T	−
6	F	T	−
7	F	F	−
8	T	F	+
9	F	T	−

Entropy Formula

$$Entropy(S) = -p_+ \log_2(p_+) - p_- \log_2(p_-)$$

Where:

- $p_+ = \frac{+}{total}$
- $p_- = \frac{-}{total}$

Entropy for a2

Entropy for a2 = T (2+, 3−)
Total = 5
Probabilities:

$$p_+ = \frac{2}{5} = 0.4$$
$$p_- = \frac{3}{5} = 0.6$$

Entropy for a2

Apply formula:

$$Entropy(T) = -0.4\log_2(0.4) - 0.6\log_2(0.6)$$

Now approximate:

- $\log_2(0.4) \approx -1.32$
- $\log_2(0.6) \approx -0.737$

Entropy for a2

Calculation:

$$= -(0.4)(-1.32) - (0.6)(-0.737)$$

$$= 0.528 + 0.442$$

$$\approx 0.97$$

Entropy for a2 = F (2+, 2−)

Total = 4

Probabilities:

$$p_+ = \frac{2}{4} = 0.5$$

$$p_- = \frac{2}{4} = 0.5$$

Entropy for a2 = F (2+, 2–)

Apply formula:

$$Entropy(F) = -0.5\log_2(0.5) - 0.5\log_2(0.5)$$

We know:

$$\log_2(0.5) = -1$$

Entropy for a2 = F (2+, 2−)

Calculation:

$$= -(0.5)(-1) - (0.5)(-1)$$

$$= 0.5 + 0.5 = 1$$

Weighted Entropy of a2

Now we combine both parts.
Total dataset = **9 instances**

Formula:

$$Entropy(a2) = \frac{5}{9} \cdot Entropy(T) + \frac{4}{9} \cdot Entropy(F)$$

Substitute values:

$$= \frac{5}{9}(0.971) + \frac{4}{9}(1.0)$$

Weighted Entropy of a2

$$Entropy(a2) = 0.539 + 0.444 = 0.983$$

$$Entropy(a2) \approx 0.984$$

Information Gain (a2)

formula

$$G(a1) = Entropy(S) - Entropy(a1)$$

Where:

- **Entropy(S)** = impurity of full dataset (before split)
- **Entropy(a2)** = weighted entropy after splitting on a1

$$IG(a2) = 0.991 - 0.984$$

IG(a2) ≈ 0.007

Which is the best splitting attribute between a1 and a2.

Attribute	Information Gain
a1	**0.230 (BEST)**
a2	0.007

Root node = a1 (best split)
because it has the **highest Information Gain**

Example 2:

ID	Study Hours	Attendance	Result
1	High	Good	Pass
2	High	Poor	Pass
3	Medium	Good	Pass
4	Medium	Poor	Fail
5	Low	Good	Fail
6	Low	Poor	Fail
7	Medium	Good	Pass
8	High	Good	Pass

Target Variable

- Target = **Result (Pass / Fail)**

ID	Study Hours	Attendance	Result
1	High	Good	Pass
2	High	Poor	Pass
3	Medium	Good	Pass
4	Medium	Poor	Fail
5	Low	Good	Fail
6	Low	Poor	Fail
7	Medium	Good	Pass
8	High	Good	Pass

$\boldsymbol{Entropy(S)}$

Total = 8
Pass = 5
Fail = 3

$$Entropy(S) = -\left(\frac{5}{8}\log_2\frac{5}{8} + \frac{3}{8}\log_2\frac{3}{8}\right)$$

$$\boldsymbol{Entropy(S) \approx 0.954}$$

Try Splitting Attributes

We check:
- Study Hours
- Attendance

ID	Study Hours	Attendance	Result
1	High	Good	Pass
2	High	Poor	Pass
3	Medium	Good	Pass
4	Medium	Poor	Fail
5	Low	Good	Fail
6	Low	Poor	Fail
7	Medium	Good	Pass
8	High	Good	Pass

Information Gain for Study Hours

Groups:
High:
•Pass = 3, Fail = 0 → Entropy = 0
Medium:
•Pass = 2, Fail = 1 → Entropy ≈ 0.918
Low:
•Pass = 0, Fail = 2 → Entropy = 0

ID	Study Hours	Attendance	Result
1	High	Good	Pass
2	High	Poor	Pass
3	Medium	Good	Pass
4	Medium	Poor	Fail
5	Low	Good	Fail
6	Low	Poor	Fail
7	Medium	Good	Pass
8	High	Good	Pass

Weighted Entropy:

Weighted Entropy:

$$= \frac{3}{8}(0) + \frac{3}{8}(0.918) + \frac{2}{8}(0) = 0.344$$

ID	Study Hours	Attendance	Result
1	High	Good	Pass
2	High	Poor	Pass
3	Medium	Good	Pass
4	Medium	Poor	Fail
5	Low	Good	Fail
6	Low	Poor	Fail
7	Medium	Good	Pass
8	High	Good	Pass

Information Gain:

$$IG = 0.954 - 0.344 = 0.61$$

ID	Study Hours	Attendance	Result
1	High	Good	Pass
2	High	Poor	Pass
3	Medium	Good	Pass
4	Medium	Poor	Fail
5	Low	Good	Fail
6	Low	Poor	Fail
7	Medium	Good	Pass
8	High	Good	Pass

Information Gain for Attendance

Good:
•Pass = 4, Fail = 1 → Entropy ≈ 0.722
Poor:
•Pass = 1, Fail = 2 → Entropy ≈ 0.918

ID	Study Hours	Attendance	Result
1	High	Good	Pass
2	High	Poor	Pass
3	Medium	Good	Pass
4	Medium	Poor	Fail
5	Low	Good	Fail
6	Low	Poor	Fail
7	Medium	Good	Pass
8	High	Good	Pass

Weighted Entropy:

$$= \frac{5}{8}(0.722) + \frac{3}{8}(0.918) = 0.796$$

ID	Study Hours	Attendance	Result
1	High	Good	Pass
2	High	Poor	Pass
3	Medium	Good	Pass
4	Medium	Poor	Fail
5	Low	Good	Fail
6	Low	Poor	Fail
7	Medium	Good	Pass
8	High	Good	Pass

Information Gain:

$$IG = 0.954 - 0.796 = 0.158$$

ID	Study Hours	Attendance	Result
1	High	Good	Pass
2	High	Poor	Pass
3	Medium	Good	Pass
4	Medium	Poor	Fail
5	Low	Good	Fail
6	Low	Poor	Fail
7	Medium	Good	Pass
8	High	Good	Pass

Best Attribute Selection

Attribute	Information Gain
Study Hours	**0.61 (BEST)**
Attendance	0.158

ID	Study Hours	Attendance	Result
1	High	Good	Pass
2	High	Poor	Pass
3	Medium	Good	Pass
4	Medium	Poor	Fail
5	Low	Good	Fail
6	Low	Poor	Fail
7	Medium	Good	Pass
8	High	Good	Pass

DECISION TREE

```
[ Study Hours ]
      /    |    \
Low   Medium   High
      |      |      |
Fail  [Attendance] Pass
            /   \
        Good   Poor
          |      |
        Pass   Fail
```

ID	Study Hours	Attendance	Result
1	High	Good	Pass
2	High	Poor	Pass
3	Medium	Good	Pass
4	Medium	Poor	Fail
5	Low	Good	Fail
6	Low	Poor	Fail
7	Medium	Good	Pass
8	High	Good	Pass

IF Study Hours = Low → **Fail**
IF Study Hours = High → **Pass**
IF Study Hours = Medium AND Attendance = Good → **Pass**
IF Study Hours = Medium AND Attendance = Poor → **Fail**

ID	Study Hours	Attendance	Result
1	High	Good	Pass
2	High	Poor	Pass
3	Medium	Good	Pass
4	Medium	Poor	Fail
5	Low	Good	Fail
6	Low	Poor	Fail
7	Medium	Good	Pass
8	High	Good	Pass